

Clinical Study Summary Report (CSSR)

Document Status: Final | Version: 1.0 | Date: 01-APR-2026

1. Administrative & Study Identification

Full Study Title:	Phase 1 Trial of MK-2870 as Monotherapy and in Combination With Pembrolizumab ± Chemotherapy in Subjects With Advanced Solid Tumors
Short Title:	Sacituzumab Tirumotecan (MK-2870) as Monotherapy and in Combination With Pembrolizumab in Participants With Advanced Solid Tumors (MK-2870-008)
Protocol Number:	309937517547042970
NCT Number:	NCT06049212
Sponsor Name:	Merck
Investigational Product:	pembrolizumab
Phase:	PHASE1
Medical Monitor:	[Not Provided in Posting]

2. Study Synopsis & Rationale

2.1 Study Objective

Primary Objective:	1. Number of Participants Experiencing Dose-Limiting Toxicity (DLT): A DLT is defined as any of the following: Grade 4 nonhematologic toxicity; Grade 4 hematologic toxicity lasting ≥7 days, except thrombocytopenia or lymphopenia; any nonhematologic AE ≥Grade 3 in severity (with some exceptions); any Grade 3 or Grade 4 nonhematologic laboratory value (if certain criteria are met); febrile neutropenia Grade 3 or Grade 4; a prolonged delay (≥2 weeks) in initiating sacituzumab tirumotecan second dose due to intervention-related toxicity; any intervention-related toxicity that causes the participant to discontinue intervention during DLT evaluation period; or any Grade 5 toxicity. All toxicities will be graded using National Cancer Institute Common Terminology Criteria for Adverse Events (NCI-CTCAE) v5.0. 2. Number of Participants Experiencing ≥1 Adverse Event (AE): An AE is any untoward medical occurrence in a clinical study participant, temporally associated with the use of study intervention, whether or not considered related to the study intervention. 3. Number of Participants Discontinuing from Study Therapy due to AE: An AE is any untoward medical occurrence in a clinical study participant, temporally associated with the use of study intervention, whether or not considered related to the study intervention.
Secondary Obj.:	1. Area Under the Plasma Concentration-Time Curve (AUC) of sacituzumab tirumotecan -Antibody-Drug Conjugate (sacituzumab tirumotecan ADC) 2. Maximum Plasma Concentration (Cmax) of sacituzumab tirumotecan-ADC 3. Minimum Plasma Concentration (Cmin) of sacituzumab tirumotecan-ADC

2.2 Background & Rationale

This is a phase 1 trial of the safety, tolerability, and pharmacokinetics (PK) of sacituzumab tirumotecan monotherapy, and of sacituzumab tirumotecan in combination with pembrolizumab (MK-3475) or pembrolizumab + carboplatin, in

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Japanese participants with advanced solid tumors or treatment-naïve advanced or metastatic non-small cell lung cancer (NSCLC).

3. Study Design & Methodology

Design Framework: Type: INTERVENTIONAL, Phase: PHASE1, Status: RECRUITING

Target N: 30

4. Subject Selection (Eligibility Criteria)

4.1 Inclusion Criteria

- * Arm 1: Histologically or cytologically confirmed advanced/metastatic solid tumor by pathology report and have received, or been intolerant to, all treatment known to confer clinical benefit.
- * Arm 2: Have a histologically or cytologically confirmed diagnosis of advanced or metastatic NSCLC (Stage IIIB or IIIC disease and not candidates for surgical resection or definitive chemoradiation, or Stage IV NSCLC, AJCC Staging Manual, version 8).
- * Arm 2: Confirmation that EGFR-, ALK-, or ROS1-directed therapy is not indicated as primary therapy
- * Arm 2: Has tumor tissue that demonstrates PD-L1 TPS ≥ 50% as determined by PD-L1 IHC 22C3 pharmDx assay by local laboratory
- * If capable of producing sperm, the participant agrees to the following during the intervention period and for at least the time needed to eliminate each study intervention after the last dose of study intervention (100 days for sacituzumab tirumotecan and 90 days for carboplatin \[no restriction for pembrolizumab\]) AND agrees to refrain from donating sperm AND is either abstinent and agrees to remain abstinent or uses highly effective contraception
- * For females (assigned at birth), is not pregnant or breastfeeding and ≥1 of the following applies: is not a participant of childbearing potential (POCBP) OR is a POCP and uses highly effective contraception
- * Arm 1: Participants who have AEs due to previous anticancer therapies must have recovered to ≤Grade 1 or baseline
- * Measurable disease by RECIST 1.1 as assessed by the local site investigator/radiology
- * Archival tumor tissue sample or newly obtained core or incisional biopsy of a tumor lesion not previously irradiated has been provided
- * Have a life expectancy of at least 3 months
- * Have an ECOG performance status of 0 or 1 within 3 days before the start of study intervention

4.2 Exclusion Criteria

- * Has symptomatic ascites or pleural effusion. A participant who is clinically stable following treatment for these conditions (including therapeutic thoraco- or paracentesis) is eligible
- * Has Grade ≥2 peripheral neuropathy
- * Has history of documented severe dry eye syndrome, severe Meibomian gland disease and/or blepharitis, or severe corneal disease that prevents/delays corneal healing
- * Has active inflammatory bowel disease requiring immunosuppressive medication or previous history of inflammatory bowel disease
- * Has uncontrolled, significant cardiovascular disease or cerebrovascular disease including New York Heart Association Class III or IV congestive heart failure, unstable angina, myocardial infarction, uncontrolled symptomatic arrhythmia,

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prolongation of QTcF interval to ≥ 480 ms, and/or other serious cardiovascular and cerebrovascular diseases within the 6 months preceding study intervention

- * Received prior treatment with a TROP2-targeted ADC
- * Received prior treatment with a topoisomerase I-containing ADC
- * Arm 1: Has received any prior immunotherapy and was discontinued from that treatment due to a Grade 3 or higher irAE (except endocrine disorders that can be treated with replacement therapy) or was discontinued from that treatment due to Grade 2 myocarditis or recurrent Grade 2 pneumonitis
- * Arm 2: Received prior therapy with an anti-PD-1, anti-PD-L1, or anti-PD-L2 agent or with an agent directed to another stimulatory or coinhibitory T-cell receptor (eg, CTLA-4, OX-40, CD137)
- * Arm 2: Has received prior systemic chemotherapy or other targeted or biological antineoplastic therapy for their advanced or metastatic NSCLC
- * Arm 2: Has received radiation therapy to the lung that is ≥ 30 Gy within 6 months of the first dose of study intervention
- * Received prior systemic anticancer therapy including investigational agents within 4 weeks before allocation
- * Received prior radiotherapy within 2 weeks of start of study intervention, or radiation-related toxicities, requiring corticosteroids
- * Received a live or live-attenuated vaccine within 30 days before the first dose of study intervention
- * Requires treatment with a strong inhibitor or inducer of CYP3A4 at least 14 days before the first dose of study intervention and throughout the study
- * Has received an investigational agent or has used an investigational device within 4 weeks prior to study intervention administration
- * Arm 2: Diagnosis of immunodeficiency or is receiving chronic systemic steroid therapy (in dosing exceeding 10 mg daily of prednisone equivalent) or any other form of immunosuppressive therapy within 7 days prior the first dose of study medication
- * Arm 2: Active autoimmune disease that has required systemic treatment in the past 2 years except replacement therapy (eg, thyroxine, insulin, or physiologic corticosteroid)
- * Known additional malignancy that is progressing or has required active treatment within the past 3 years
- * Known active CNS metastases and/or carcinomatous meningitis
- * History of (noninfectious) pneumonitis/interstitial lung disease that required steroids or has current pneumonitis/interstitial lung disease
- * Active infection requiring systemic therapy
- * History of HIV infection
- * Concurrent active Hepatitis B (defined as HBsAg positive and/or detectable HBV DNA) and Hepatitis C virus (defined as anti-HCV Ab positive and detectable HCV RNA) infection
- * History or current evidence of any condition, therapy, laboratory abnormality, or other circumstance that might confound the results of the study or interfere with the participant's participation for the full duration of the study, such that it is not in the best interest of the participant to participate, in the opinion of the treating investigator
- * Known psychiatric or substance abuse disorder
- * A severe hypersensitivity reaction to treatment a monoclonal antibody/component of the study intervention

11. Study Identifiers & Governance

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Registry Number:	NCT06049212
Sponsor:	Merck
Study Title:	Sacituzumab Tirumotecan (MK-2870) as Monotherapy and in Combination With Pembrolizumab in Participants With Advanced Solid Tumors (MK-2870-008)

15. Sign-off & Approval

Principal Investigator: _____ Date: _____
Clinical Operations Lead: _____ Date: _____
Lead Statistician: _____ Date: _____